

Indian Statistical Institute

BSDS - Second Year

Exercise Set 4.

1. Let $\{S_n : n \geq 0\}$ be the simple symmetric ($p = 1/2$) random walk in one dimension with $S_0 = 0$.

- a) Show that $\{|S_n| : n \geq 0\}$ a Markov chain on $\{0, 1, 2, \dots\}$. Find its transition matrix.
- b) Let $M_n = \max\{S_k : 0 \leq k \leq n\}$, and show that $Y_n = M_n - S_n$ defines a Markov chain. Find the transition probabilities of this chain.

What happens if the random walk is simple but asymmetric? What happens if the starting position is not the origin, i.e., $S_0 \neq 0$.

2. A die is rolled repeatedly. The results be denoted by $\{\xi_n : n \geq 1\}$. Which of the following are Markov chains? For those that are, find the transition matrix.

- a) The largest number $\{M_n : n \geq 0\}$ shown up to the $(n + 1)$ -th roll, i.e., $M_n = \max\{\xi_1, \dots, \xi_{n+1}\}$.
- b) The number N_n of sixes in n rolls, i.e., $N_n = \sum_{k=1}^n \mathbb{I}(\epsilon_{k+1} = 6)$.
- c) The number of throws since the last six, i.e., $L_n = n - \max\{k \leq n : \epsilon_{k+1} = 6\}$, with the convention that if the set $\{k \leq n : \epsilon_{k+1} = 6\}$ is empty, the maximum is taken to be 0.
- d) The number of throws until the next six, $R_n = \min\{k > n : \epsilon_{k+1} = 6\} - n$.

3. Suppose that $\{\xi_n : n \geq 1\}$ denote the scores obtained at the n -th throw of a fair dice. Set $S_n = (\epsilon_{n+1} + \epsilon_{n+2}) \bmod 6$ for $n \geq 0$. Show that $\{S_n : n \geq 1\}$ is a Markov Chain. Find the transition probability matrix for this chain.

4. Let $\{X_n : n \geq 0\}$ be a Markov chain on the state space $\mathcal{S}$ and transition matrix $\mathbf{P}$. Set, for $k \geq 1$ and $n \geq 0$,

$$Y_n := (X_n, X_{n+1}, \dots, X_{n+k}). \quad (1)$$

Show that the process $\{Y_n : n \geq 0\}$, taking values $\mathcal{S}^{k+1}$, is a Markov chain. What is its transition matrix? The Markov chain $\{Y_n : n \geq 0\}$ is often referred as the snake chain of length $k + 1$ associated with $\{X_n : n \geq 0\}$.

5. Consider the chain $\{X_n : n \geq 0\}$, where X_n 's represents the result of independent tosses of a coin with probability of success being p . For $k = 2$, we sample space consists of all possible patterns of length 3, i.e., $\{(H,H,H), (H,H,T), (H,T,H), (H,T,T), (T,H,H), (T,H,T), (T,T,H), (T,T,T)\}$ (we will drop the separating comma in future). Find the transition matrix this snake chain.

6. Let $\{X_n^{(j)} : n \geq 0\}$ be independent sequences of Markov chains for $1 \leq j \leq k$ on the state space $\mathcal{S}_j$ and transition matrix $\mathbf{P}_j$. Set, for $n \geq 0$,

$$Y_n := (X_n^{(1)}, X_n^{(2)}, \dots, X_n^{(k)}). \quad (2)$$

Show that the process $\{Y_n : n \geq 0\}$, taking values $\prod_{j=1}^k \mathcal{S}_j$, is a Markov chain. What is its transition matrix?

7. Suppose that a rook and a bishop has been placed at opposite corners of an otherwise vacant chessboard. The pieces move simultaneously and independently of each other to squares where they can legally move. Note that we are allowing the pieces to come to the same square at the same time. Formulate this as a Markov chain. What is its transition matrix?

8. Let $\{\xi_n : n \geq 0\}$ be a sequence of independent identically distributed random variables taking values in $\{-1, 1\}$ with $\mathbb{P}(\xi_n = 1) = p = 1 - \mathbb{P}(\xi_n = -1)$. Let $X_n = \xi_n \xi_{n+1}$, for all $n \geq 0$. Is $\{X_n : n \geq 0\}$ a Markov chain?

9. Let $\{\xi_n : n \geq 1\}$ be a sequence of independent identically distributed integer-valued random variables. Let $S_0 = 0$ and for $n \geq 1$, $S_n = \sum_{i=1}^n \xi_i$. Also define $X_0 = 0$, $X_1 = \xi_1$ and for $n \geq 2$, $X_n = \xi_n + \xi_{n-1}$ and $Y_0 = 0$ and for $n \geq 1$, $Y_n = \sum_{k=1}^n S_k$. Which of the following constitute Markov chains : (a) $\{X_n : n \geq 0\}$, (b) $\{Y_n : n \geq 0\}$ and (c) the sequence of pairs $\{(S_n, Y_n) : n \geq 0\}$?

10. Find an example of a Markov chain $\{X_n : n \geq 0\}$ on the state space $\mathcal{S}$ and $x_0, x_2 \in \mathcal{S}$ and $A_1 \subseteq \mathcal{S}$ (having at least two elements) such that

$$\mathbb{P}(X_2 = x_2 \mid X_1 \in A_1, X_0 = x_0) \neq \mathbb{P}(X_2 = x_2 \mid X_1 \in A_1).$$

11. Consider the chain defined in Exercise 5. Set $A_0 = \{HHT, HTT, THT, TTT\}$, $A_1 = \{HTH, TTH\}$, $A_2 = \{THH\}$ and $A_3 = \{HHH\}$. We define a new process, called *aggregated process*, $\{Y_n : n \geq 0\}$ as follows: for any $i \in \{0, 1, 2, 3\}$, define $Y_n = i$ if $X_n \in A_i$. Show that the aggregated process is a Markov chain on $\{0, 1, 2, 3\}$. Find its transition matrix.

12. Show that not all stochastic matrices can be achieved as $\mathbf{P}^2$.

13. Set $Z_0 = 0$ and $Z_n = \max\{X_1, X_2, \dots, X_n\}$ for $n \geq 1$ where X_i are **iid** random variables taking values in $\{0, 1, 2, \dots\}$. Show that $\{Z_n : n \geq 1\}$ is a Markov chain. Find $p_{ij}^{(n)}$.

14. Consider a branching process $\{Z_n\}$. Suppose the offspring distribution is $\mathbb{P}(\xi = 0) = 0.2, \mathbb{P}(\xi = 1) = 0.5, \mathbb{P}(\xi = 2) = 0.3$.

1. Compute the transition probabilities p_{0j}, p_{1j}, p_{2j} for $j = 0, 1, 2, 3, 4$.
2. If the population starts with a single individual ($Z_0 = 1$), what is the probability that the population is extinct ($Z_1 = 0$) after one generation?
3. What is the probability that the population has exactly 2 individuals after one generation?

15. Consider the chain defined in Exercise 5. Compute $p_{ij}^{(n)}$. Consider the aggregated chain defined in Exercise 11. Can you find $p_{ij}^{(n)}$?